

P12B Macro Project: Forward Brainstorm (Option Map, Not Fixed Plan)

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1 Decision Rule (Hard Filter)

Continue only if an option satisfies both:

1. Strong portfolio signal for applied economist / economic data analyst roles.
2. Realistic delivery in 14 days or less with stable execution.

If either fails, stop or defer that branch.

2 Branch Map: All Meaningful Directions

2.1 A. Identification and Causal Inference Branches

A1. Country and year fixed effects (two-way FE)

- Goal: absorb time-invariant country heterogeneity and global shocks.
- Value: standard baseline for panel credibility.
- Risk: still vulnerable to time-varying omitted variables and simultaneity.

A2. FE plus controls

- Add controls such as trade openness, fiscal proxy, financial depth, exchange-rate regime proxy if available.
- Value: better model defensibility in interviews.
- Risk: data collection burden and model instability if controls are weak/incomplete.

A3. IV / 2SLS branch

- Candidate idea set: external monetary pressure proxies, lagged external instruments, commodity-linked external shocks for suitable subsamples.
- Mandatory diagnostics: first-stage F, weak-IV robust inference, over-identification checks (if over-identified).
- Value: highest signal if instrument story is defensible.

- Risk: instrument validity is the main failure point.

A4. Dynamic panel branch

- Add lagged dependent variables and dynamic adjustment story.
- Value: closer to macro transmission logic.
- Risk: complexity rises quickly; easy to over-scope beyond 14 days.

2.2 B. Robustness and Stress-Test Branches

B1. Alternative variable definitions

- Money proxy variants (M2 level growth vs ratio transformations).
- Inflation proxy variants (CPI-based alternatives where feasible).
- Value: demonstrates that findings are not a single-definition artifact.

B2. Sample slicing

- By period (pre/post major global events), income groups, inflation regimes.
- Value: strengthens external validity narrative.
- Risk: smaller cells reduce statistical power.

B3. Outlier influence checks

- Winsorization / robust regression / leave-one-country-out checks.
- Value: protects against “single-country drives everything” critique.

2.3 C. Portfolio Productization Branches

C1. Reproducible research package

- One canonical notebook, one exported dataset, one methods note, one results note.
- Value: immediate hiring signal for execution quality.

C2. Interview-facing one-pager

- Problem, method choice, key result, caveat, business relevance.
- Value: high communication ROI.

C3. Light dashboard/static chart sheet

- Non-interactive static figures acceptable if consistent and clean.
- Value: faster than full app build; avoids over-engineering.

2.4 D. Adjacent Extension Branches

D1. Forecasting extension

- Use money-growth information as one feature in inflation forecasting benchmarks.
- Value: links causal and predictive framing.
- Risk: can dilute core identification story if too broad.

D2. Policy / risk memo variant

- Convert findings into a short macro-risk interpretation memo.
- Value: direct relevance for risk and analytics roles.

3 Portfolio Value Matrix (Initial Scoring)

Scale: High / Medium / Low.

Option	Portfolio Signal	Causal Credibility	14-Day Fea-sibility	Recommended Status
Two-way FE baseline	High	Medium	High	Do first if continuing
FE + controls	High	Medium-High	Medium	Do if data coverage is acceptable
IV / 2SLS full branch	Very High	High (if valid)	Medium-Low	Do only with credible instrument story
Dynamic panel	Medium-High	Medium	Low-Medium	Optional, only after FE core is stable
Robustness pack (definitions + outliers + splits)	High	Medium-High	High	Strong 14-day candidate
Reproducible package + one-pager	Very High	N/A	Very High	Mandatory for portfolio value
Forecasting extension	Medium	Low-Medium	Medium	Secondary branch
Policy/risk memo	High	N/A	High	Strong add-on after core results

4 Practical Continue / Stop Triggers

Continue if within first 3 work sessions you can produce:

- A stable two-way FE result with interpretable sign/magnitude.
- At least two robustness checks that do not reverse the baseline story completely.

- One concise methods/results writeup usable in interview discussion.

Stop or pivot if any occurs:

- Instrument idea cannot be defended after quick validity check.
- Data merges for controls become a multiday bottleneck.
- Scope grows beyond 14-day cap without clear incremental portfolio gain.

5 Minimal High-Value Path (if continuing)

This is not a fixed plan, but the highest expected-value branch under your filter:

1. Two-way FE baseline.
2. Robustness pack (2 to 4 checks).
3. Reproducible package plus one-page narrative.
4. Optional IV add-on only if instrument quality survives fast diagnostics.

6 Open Idea Backlog (Uncommitted)

- Heterogeneous effects by inflation regime.
- Event-window around major monetary regime changes.
- Country-clustered standard errors vs alternative clustering structures.
- Bayesian shrinkage variant for noisy country-level estimates.
- Alternative money aggregates if data quality permits.